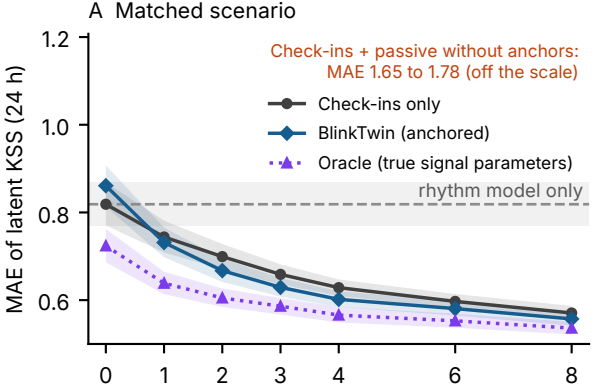
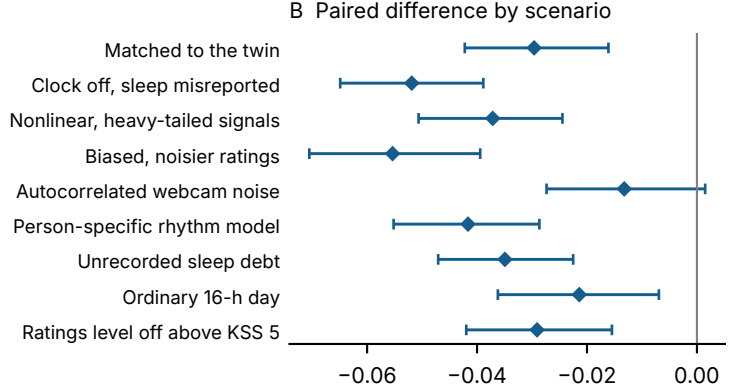

A Matched scenario
MAE of latent KSS (24 h)
1.2
1.0
0.8
0.6
Check-ins + passive without anchors:
MAE 1.65 to 1.78 (off the scale)
Check-ins only
BlinkTwin (anchored)
Oracle (true signal parameters)
rhythm model only
0
1
2
3
4
6
8
KSS check-ins in the first 16 h


B Paired difference by scenario
Matched to the twin
Clock off, sleep misreported
Nonlinear, heavy-tailed signals
Biased, noisier ratings
Autocorrelated webcam noise
Person-specific rhythm model
Unrecorded sleep debt
Ordinary 16-h day
Ratings level off above KSS 5
−0.06
−0.04
−0.02
0.00
MAE difference, BlinkTwin minus check-ins only
(k = 3; negative favours BlinkTwin; 95% CI)